

SARVESH GANESAN

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PROFESSIONAL SUMMARY

Lead AI Architect with 2.5+ years of experience building and shipping a cloud-native enterprise data platform from the ground up. Major architect of 12+ production microservices spanning agentic AI, distributed ETL, data lakehousing, and MLOps, all running on AWS EKS. Proven track record of turning ambiguous product requirements into scalable, cost-optimized systems that serve real enterprise workloads.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, FastAPI, Flask, REST APIs, gRPC, SSE streaming
- **AI / LLM:** AWS Bedrock, Strands Agents, LangChain, LangGraph, RAG systems, MCP
- **Data Engineering:** Apache Spark, Apache Iceberg, DuckDB, Pandas, PyArrow, Polars
- **Databases:** PostgreSQL, pgvector, Redis, Pinecone, MongoDB, Elasticsearch, DynamoDB
- **Cloud & Infra:** AWS (EKS, Bedrock, S3, Lambda, ECR, CloudWatch), Docker, Kubernetes, Karpenter
- **MLOps:** MLflow, Jupyter, PyTorch, TensorFlow, scikit-learn, Docker-based GPU runtimes

EXPERIENCE

Lead AI Architect Sep 2023 - Mar 2026
Groundzero Software Private Limited Chennai, India

- Owned the entire backend architecture end-to-end, designing and shipping 12+ production microservices for agentic chat, data providers, ETL orchestration, Iceberg lakehouse APIs, MLOps, and Spark/DuckDB runtimes as the sole backend engineer.
- Built a multi-agent AI system that lets enterprise users query databases, build dashboards, run ETL jobs, and manage notebooks through natural language, reducing time-to-insight from hours of manual work to seconds of conversation.
- Scaled data connectivity to 69+ sources through a dynamically loaded connector architecture with role-based authorization and schema introspection caching.
- Reduced RAG query latency by 90% using hybrid vector plus full-text retrieval with neural reranking, enabling real-time document Q&A; at production scale.
- Delivered sub-10s query performance on 100M+ row datasets by architecting a TB-scale Apache Iceberg lakehouse with Spark and DuckDB compute engines.
- Cut LLM inference costs with three-tier model routing, Anthropic prompt caching, and per-request token tracking across agent workflows.
- Designed the ETL orchestration layer for job submission, multi-job chaining, container log streaming from EKS pods, and dataset lineage across Spark, DuckDB, and Pandas engines.
- Stood up MLOps and platform infrastructure with MLflow, Jupyter lifecycle management on EKS, GPU training containers, Karpenter autoscaling, and 99.9% uptime across multi-tenant deployments.

Software Trainee Jun 2023 - Aug 2023
SCI-BI Software Solutions Private Limited Chennai, India

- Built client-facing dashboards and reports in Power BI and Tableau, and translated client requirements into technical specifications.

PROJECTS

SRE Tea MCP Server | Node.js, Model Context Protocol, Claude, JWT, AWS Lambda GitHub

- Built a production MCP server for live tea e-commerce administration, exposing 46 tools across orders, products, customers, inventory, invoices, logistics, churn analytics, and reporting with automatic token refresh and parallel API aggregation.
- Integrated Claude Desktop, Claude Code, and custom MCP clients over STDIO transport with secure environment-based credentials, structured stderr logging, and automatic recovery from expired JWT sessions.

Qualipilot Data Quality Checker | Python, Polars, Dask, cuDF, AWS Bedrock GitHub

- Built a production data-quality CLI/library with missing, duplicate, type, range, outlier, cardinality, and freshness checks, CI severity gates, HTML/JSON/Markdown reports, and optional LLM narration through Bedrock, Ollama, or OpenAI-compatible endpoints.
- Implemented swappable Polars, Pandas, Dask, and cuDF engines with typed Pydantic results, deterministic JSON output, Docker/Lambda deployment paths, and probabilistic record linkage for deduplication.

PlannerAgent | LangGraph, Anthropic, AWS Bedrock, Python GitHub

- Built a conversational planning agent supporting 10+ turn dialogues, clarifying questions, persistent compressed sessions, versioned plans, 12+ templates, Gantt charts, and Trello/CSV/Markdown import/export.
- Implemented explicit LangGraph state flow, context compaction near token limits, pickle+gzip session storage, undo/redo history, provider switching across Anthropic and Bedrock, and 40+ tests.

EDUCATION

Bachelor of Technology: Computer Science and Engineering May 2023
Specialization in Artificial Intelligence and Machine Learning GPA: 9.08
SRM Institute of Science and Technology, Chennai

LANGUAGES

English (Fluent) | Tamil (Native) | Telugu (Fluent) | Hindi (Intermediate)